

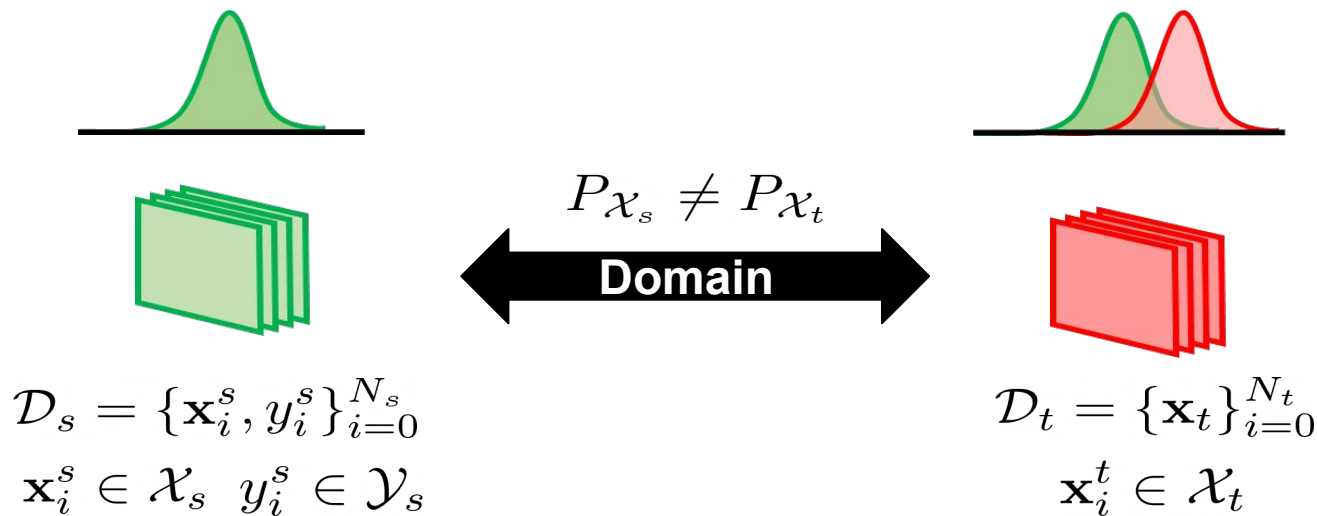
ClusT3: Information Invariant Test-Time Training

Gustavo A. Vargas Hakim*, David Osowiechi*, Mehrdad Noori, Milad Cheraghali khani,
Ali Bahri, Ismail Ben Ayed, Christian Desrosiers

*equal contribution

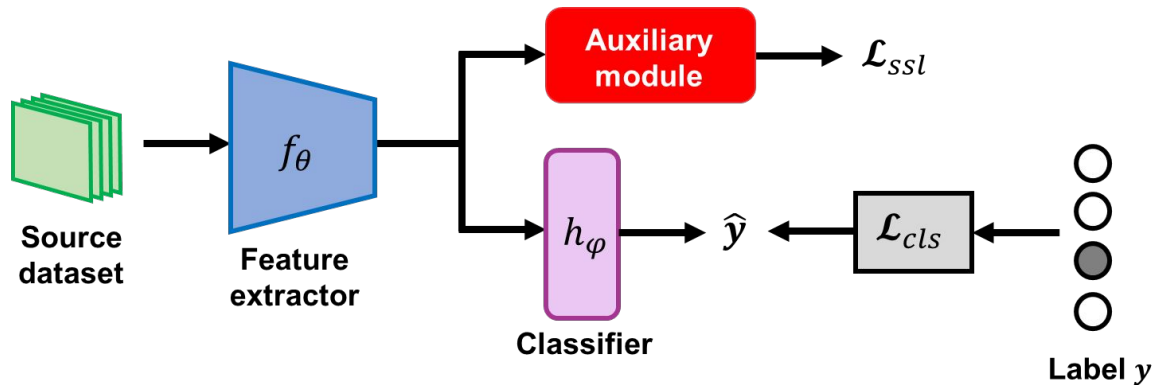
Test-Time Adaptation

In Test-Time Adaptation, a model is adapted to a target dataset without accessing the training (source) data:



Test-Time Training

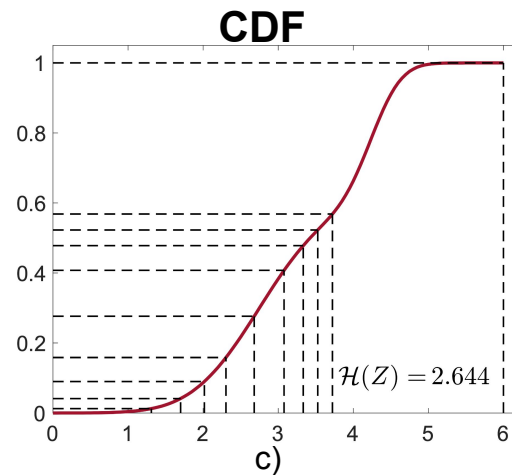
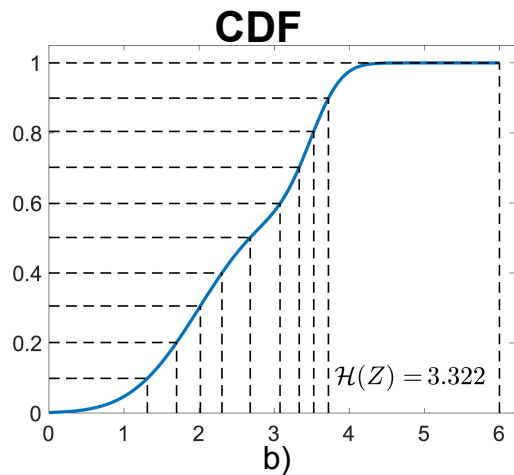
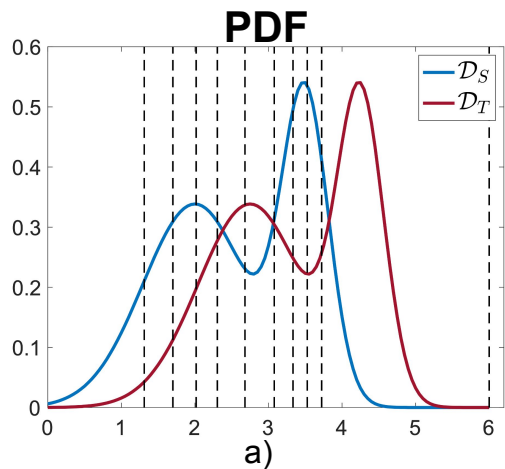
An auxiliary task is used at training time and at test-time:



Examples: TTT (rotation prediction), TTT++ (contrastive learning), TTTFlow (normalizing flow)

Information invariance between domains

Illustrative example



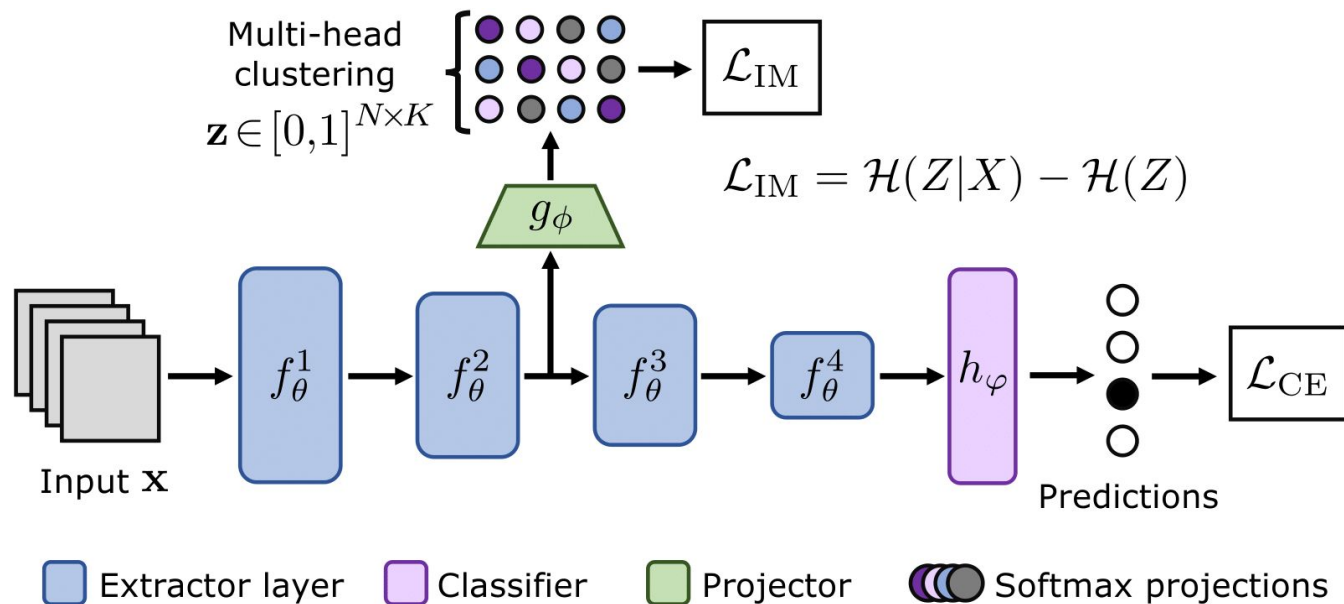
Source distribution



Target distribution

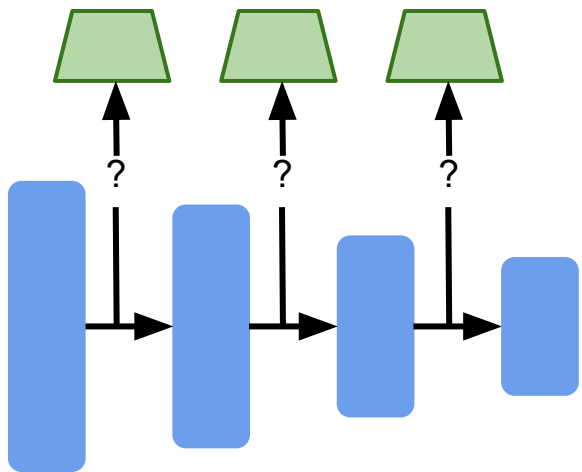
ClusT3

We maximized the **Mutual Information** between feature maps and discrete representations:



ClusT3

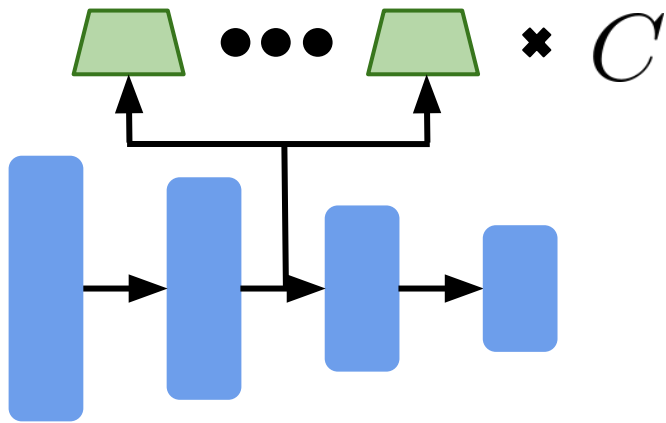
Multi-scale clustering:



$$\mathcal{L}_{CT3} = \mathcal{L}_{CE} + \sum_{\ell=j}^J \mathcal{L}_{IM}^{\ell}$$

ClusT3

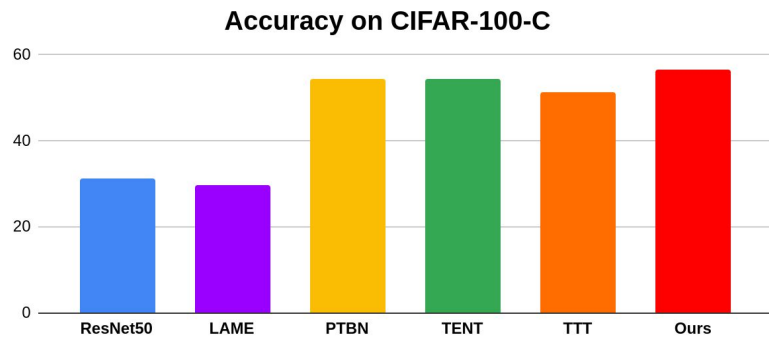
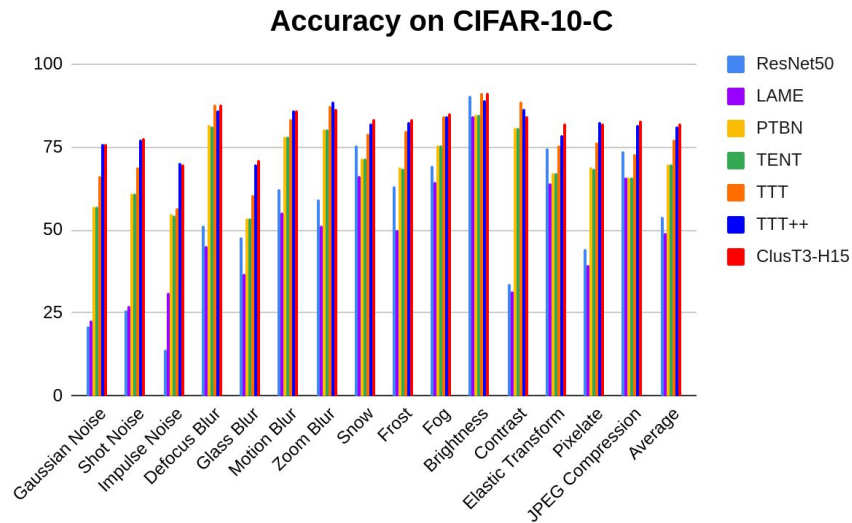
Multi-head clustering:



$$\mathcal{L}_{\text{CT3}} = \mathcal{L}_{\text{CE}} + \sum_{\ell=j}^J \sum_{\ell=c}^C \mathcal{L}_{\text{IM}}^{\ell}$$

Results: common corruptions

Using CIFAR-10/100-C: 15 different corruptions with 10K samples per corruption



Results: sym-to-real

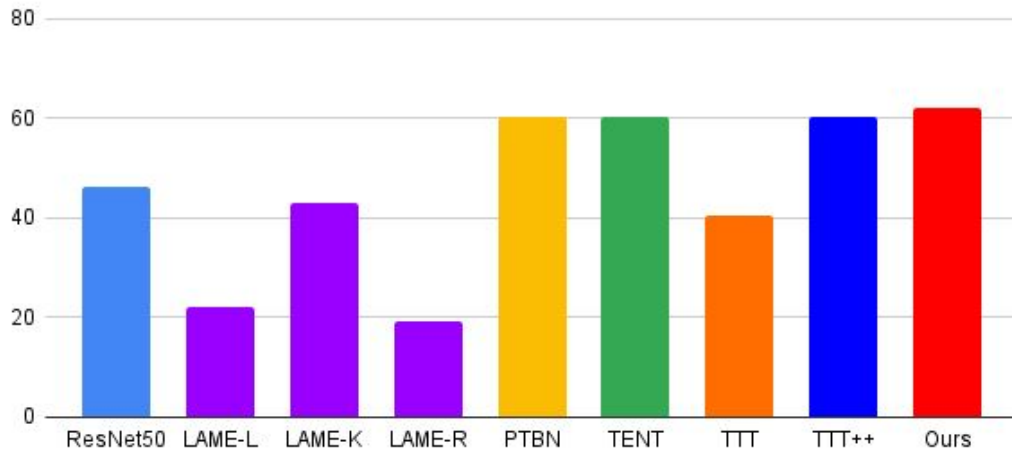
Using VisDA-C: 152,397 3D renderings → 72,372 video frames



bus

car

Accuracy on Visda-C



Contact us if you have any questions:

`gustavo-adolfo.vargas-hakim.1@ens.etsmtl.ca`

`david.osowiechi.1@ens.etsmtl.ca`

And look for the code and the whole article here:

<https://github.com/dosowiechi/ClusT3>

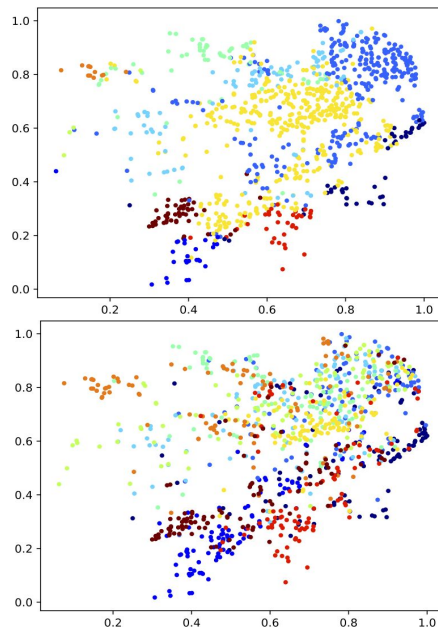


Results: common corruptions

ClusT3 leads to a more consistent clustering of feature maps:

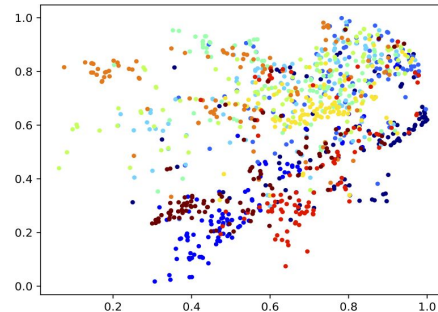
prediction **before**
adaptation

a)

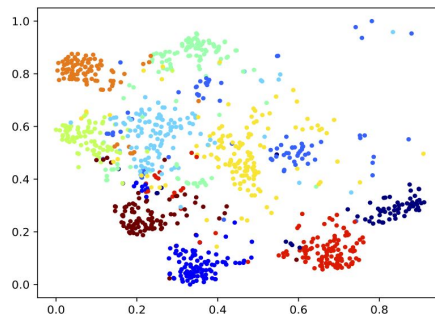


ground truth **before**
adaptation

c)



b) prediction **after**
adaptation



d) ground truth **after**
adaptation

